

JODELL Gripper AddOn

This AddOn is compatible with the JODELL RG series grippers. It includes gripper instruction blocks and a gripper control page.

Plugin Information

- Name: JODELL_Gripper_v1.1.1.tar.gz
- Version: v1.1.1

Installation Guide

System Requirements

- Controller: v1.7.1.46 or above, x64 controller
- App: cobot 2.1.10 or Zu App v1.7.1.52 or above

Installing the Plugin

Open App-Applications, or go to App Settings - System Settings - Add-ons to import and install the plugin.



Installing the Gripper

Method 1: Connect via the Controller Cabinet RS485 Channel

Electrical Connection

The JODELL RG series electric grippers are powered by DC24V and support Modbus communication control.

485 signal pin wiring definitions

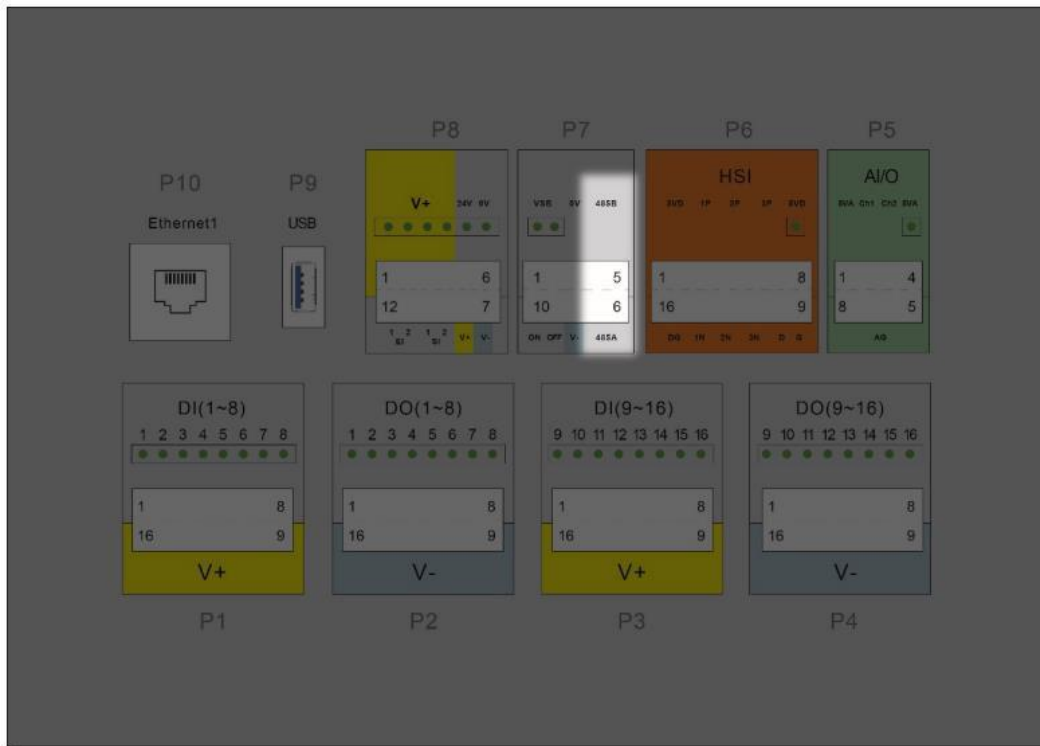
Function	Pin	Wire	Numeric Mark
485	A	White	1
485	B	Black	2

Power pin wiring definitions

Wire	Spec
Blue	DC0V
Brown	DC24V

RG52-050 IO signal pin wiring definitions

Function	Pin	Wire
IO Input	IN1	Green
IO Input	IN2	Orange
IO Output	O1	Yellow
IO Output	O2	Pink
485	A	White
485	B	Black
IO Power	DC24V	Brown
IO Power	DC0V	Blue



Note: Note: A 120Ω terminating resistor is required when wiring. YAGEO MF0207FTE52-120R is recommended.

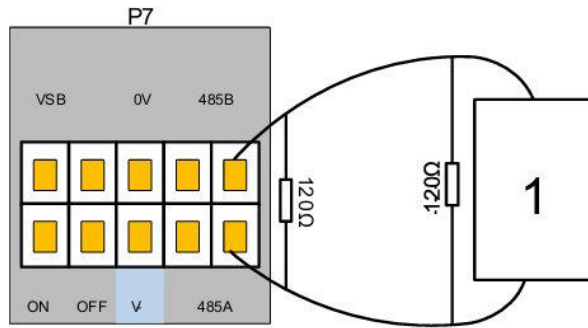
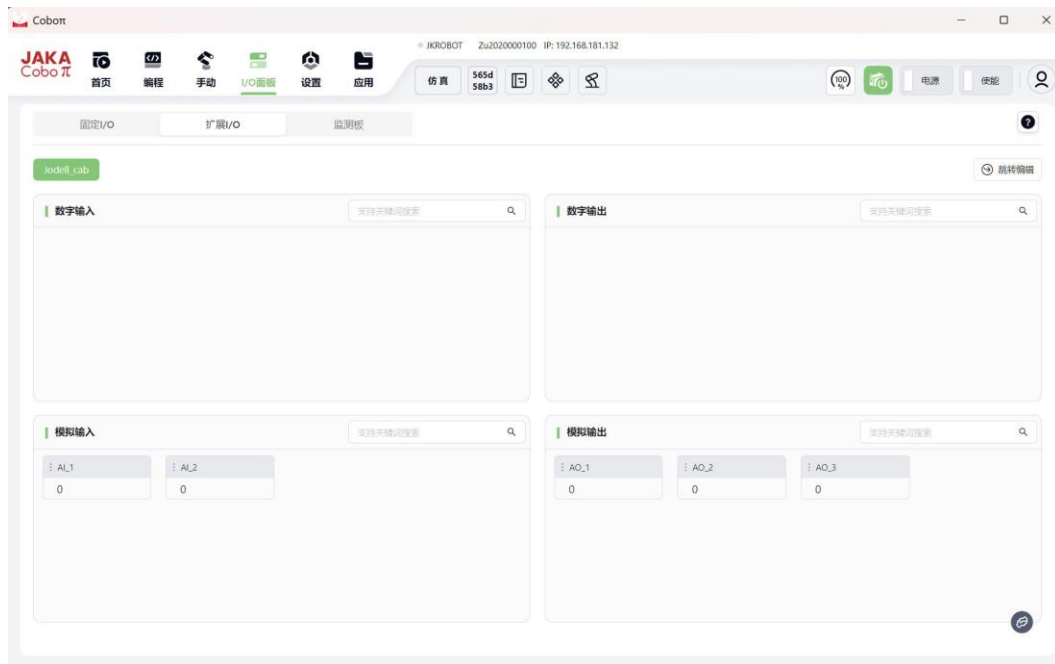
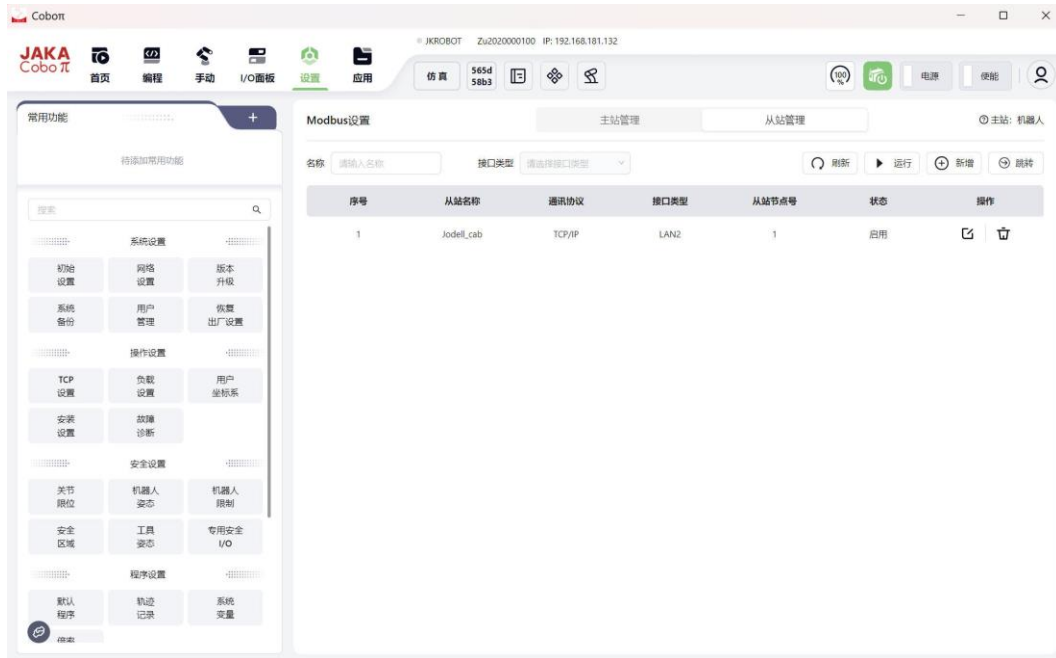


图 6-43: RS485 接口连接单个外部设备

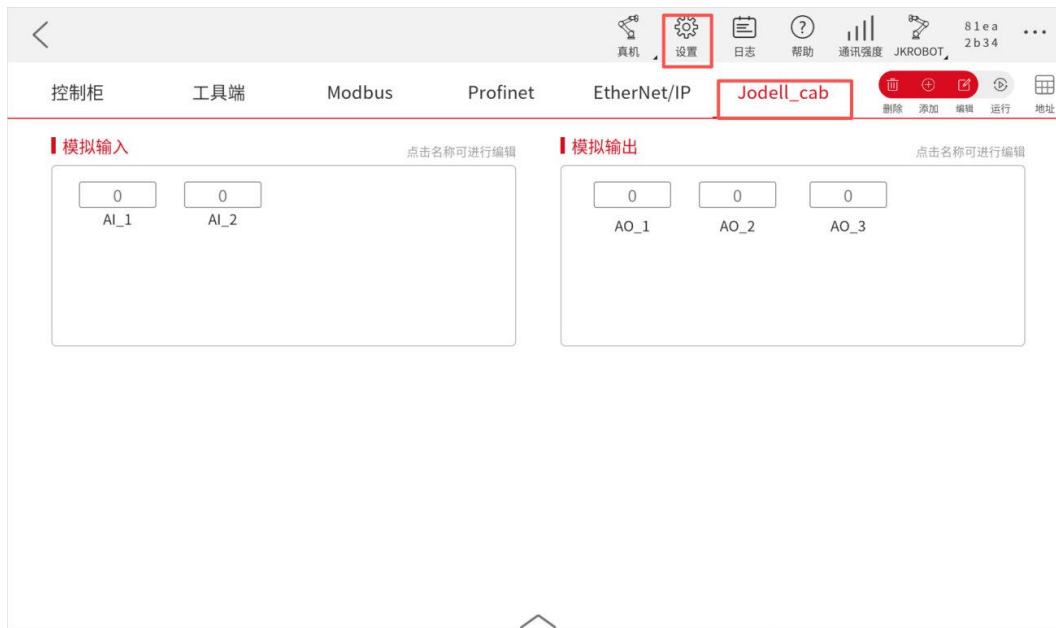
WEB App or Zu App operations:

Open WEB App - I/O Panel - Jump to edit, add extended I/O, edit and set it to Jodell_cab





Open Zu App - I/O Panel, add extended I/O, edit and set it to Jodell_cab



After configuring with the JODELL gripper's default communication parameters and address table, run the extended I/O.

I/O设置

☐ Modbus TCP/IP ☒ Modbus RTU

名称: Jodell_cab 从站节点号: 9

波特率: 115200 数据位长度: 8

停止位长度: 1 校验方式: 无校验

☐ 数字输入: 寄存器地址 数量

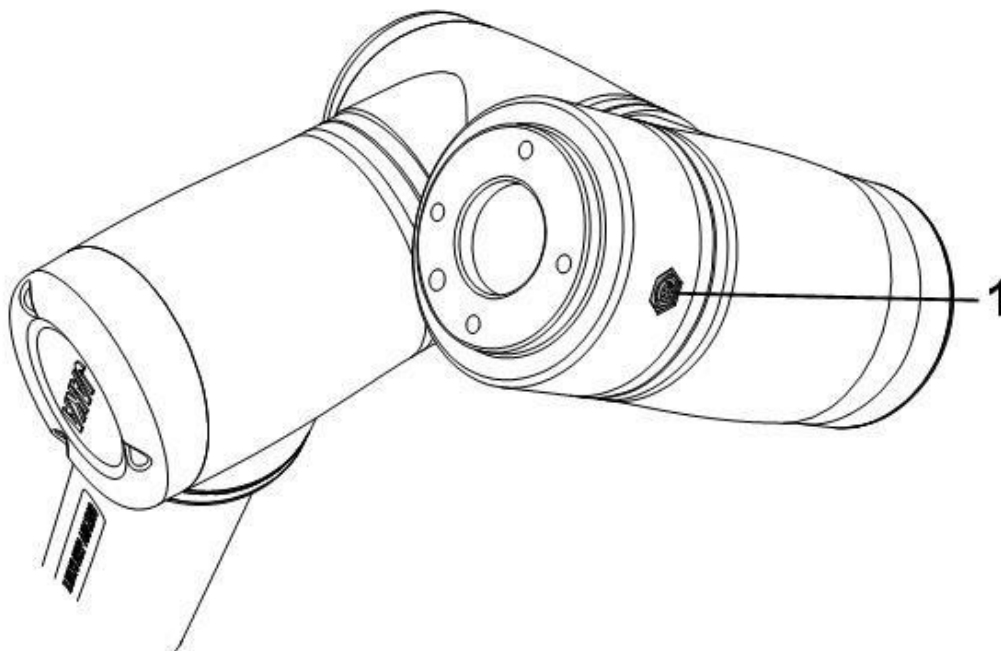
☐ 数字输出: 寄存器地址 数量

☒ 模拟输入: 寄存器地址 2000 数量 2

☒ 模拟输出: 寄存器地址 1000 数量 3

取消 确定

Method 2: Connect the Gripper to the Robot TIO Interface Using the Adapter Cable Provided by JODELL (Compatible with JAKA End TIO)







Run the plugin and click the "gear" button in the plugin operation options bar to enter the configuration page.

编辑信号量

标识符

s1

RS485通道ID

RS485通道1

类型

输入寄存器

寄存器地址

2000

刷新频率

5

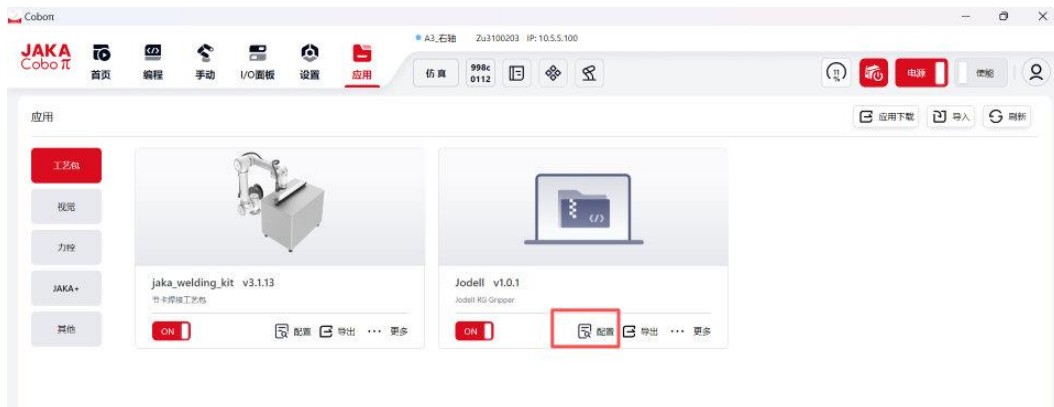
取消

确定

Step 3: Add RS485 channel 1 signals s1, s2, s3 with addresses of input registers 2000, 2001, 2002 respectively.

Main Interface

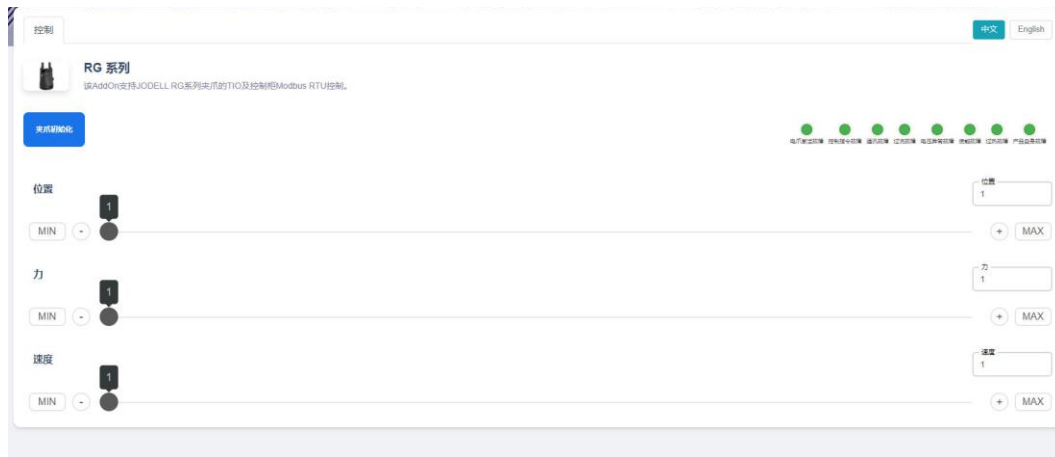
Users can control the gripper from the main interface.



Fault Information

When any indicator light turns red, the gripper is in the corresponding fault state.

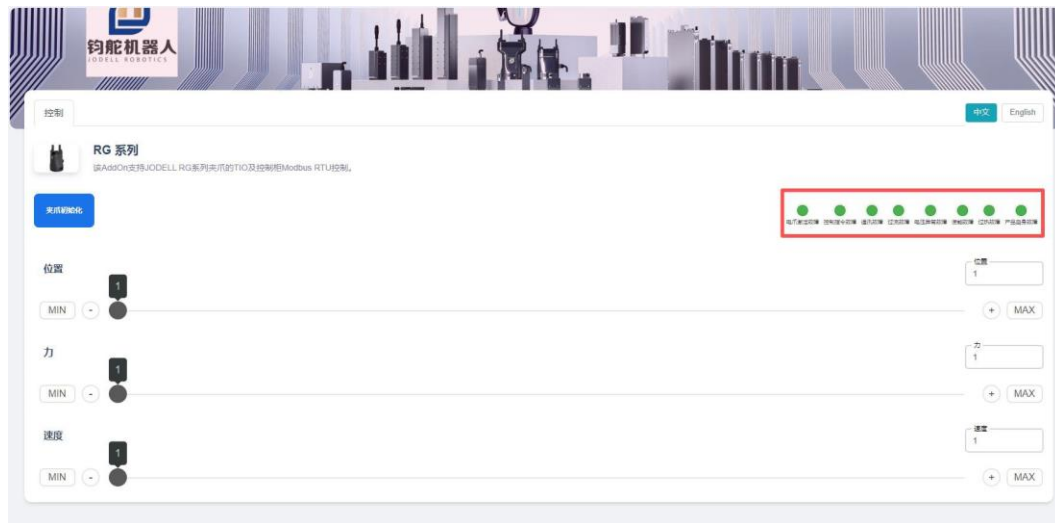
Note: Note: When the gripper is not connected, all status feedback defaults to no fault.



Usage Instructions

Instruction Blocks

Find the gripper instruction blocks in the App programming page - Extensions.



Instruction 1 - Gripper enable: Controls gripper enable and disable.



Instruction 2 - Control gripper: Supports speed (1-255), force (1-255), and position control (0-255).

Note: Note: 1. Speed, force, and position values only support positive integers. 2. When using TIO to connect the RG series high-power grippers, it is recommended to set the speed and force below 128.

Instruction 3 - Get gripper enable status: The first parameter of the dropdown is the enable status, the second parameter is the gripping status, and the third parameter is the gripper position. The gripping status values (0,1,2,3) correspond to the following:

- 0 No object detected
- 1 Object detected while fingers are opening
- 2 Object detected while fingers are closing
- 3 Fingers have reached the specified position, no object detected

Example Program

Disable the gripper first, then enable it, and move back and forth between positions 10 and 255 (open first, then close), and read the gripper status using jodell get gripper status.

The gripper performs a self-check (open and close once) during the enable process.



Note: Note: 1. When using TIO connection control, set the refresh rate of each signal during initialization. 2. When using cabinet connection, set `Jodell\_cab` as the first extended I/O module.

Troubleshooting

Common Issues

If a workpiece gets stuck or the gripper jams for other reasons during the enable process, the gripper zero position may shift. Power off and restart the gripper, then re-enable it.

Updates and Upgrades

v1.1.1

- Developed based on AddOn 3.0, supporting Modbus RTU (RS485) connection for controller cabinet 2.1 and minicab.
- Developed based on the RG series communication manual.
- Control page control support.

Support and Contact

Learn more about AddOns: <https://www.jaka.com/docs/guide/1.7.2/addOn/1.1-AboutAddOn.html>

- Contact JAKA sales or technical staff.